



# Analyzing VLM-Based Approaches for Anomaly Classification and Segmentation

**Course: Advanced Computer Vision**  
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**Presented by:**

Lawrence S Xavier Prince  
Mirudula Shri Muthukumaran  
Mohit Kakda  
Uttapreksha Patel

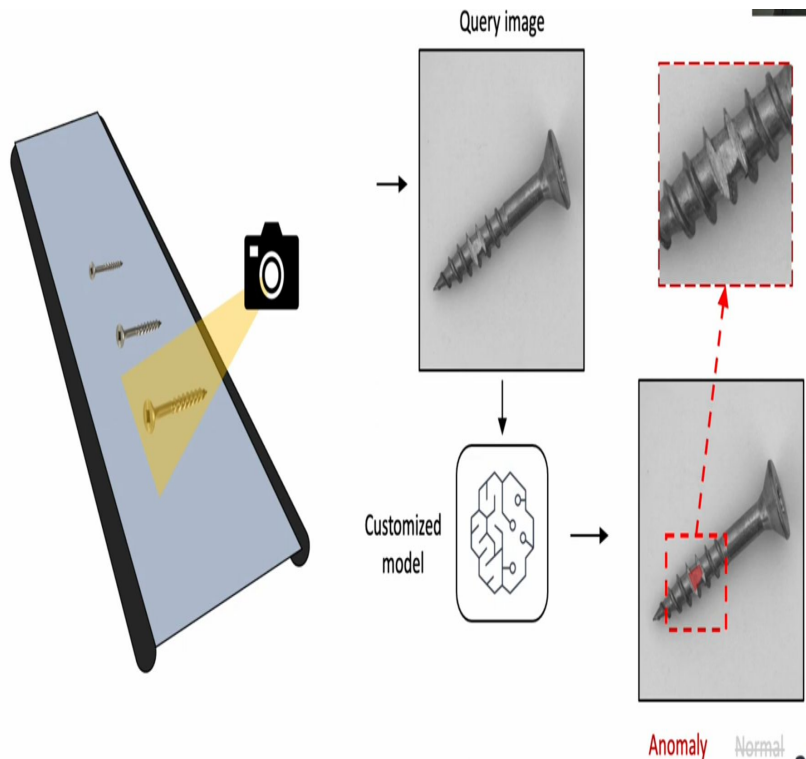
# The Problem - Industrial Quality Control

## Real-World Challenge:

- Manufacturing defects cost billions annually
- Traditional inspection: manual, slow, error-prone
- Example: Semiconductor facilities must inspect for:
  - Cracks and scratches
  - Missing components
  - Burnt traces
  - Misalignments

## The Bottleneck:

- Each new product requires retraining
- Large labeled datasets needed
- Expensive and time-consuming
- Cannot scale to new defect types quickly



# Traditional Approaches - Limitations



## Conventional Computer Vision Methods:

### 1. Task-Specific Training Required

- Extensive labeled datasets for each defect type
- Costly annotation process
- Cannot generalize to unseen defects

### 2. Major Limitations:

- New product design → Complete retraining
- New defect type → New training data required
- Production delays during model updates
- High labor costs for data annotation

**We need a better approach!**

# Our Solution - Vision-Language Models



## Key Innovation: Zero-Shot Detection

- Use pre-trained Vision-Language Models (VLMs)
- Detect defects using natural language descriptions
- No task-specific training needed

## Example:

Text Prompt: "a photo of a damaged bottle with cracks"

Model: Identifies cracks without seeing training examples

## Benefits:

- Rapid deployment to new products
- Handles unseen defect types
- Minimal data requirements
- Flexible and scalable

# Research Objectives



## Primary Goals:

1. **Systematic Analysis** of VLM-based anomaly detection methods
  - WinCLIP, AnomalyCLIP, APRIL-GAN frameworks
2. **Comparative Evaluation** across:
  - Classification accuracy (AUROC, AP)
  - Segmentation precision (pixel-AUROC, AUPRO)
  - Computational efficiency
  - Cross-domain generalization
3. **Practical Guidelines** for method selection
  - When to use which approach
  - Trade-offs and limitations
  - Real-world deployment considerations

# Evolution of Anomaly Detection



## Paradigm Shift Over Time:

### 1. Reconstruction-Based (2010s)

- Autoencoders, GANs
- Problem: Generalize too well to anomalies
- DRAEM: Synthetic anomaly generation

### 2. Embedding-Based (2019-2021)

- PatchCore: 98.2% pixel-AUROC
- SPADE, PaDiM: Statistical feature modeling
- Limitation: Need hundreds of normal samples

### 3. Self-Supervised (2021-2022)

- CutPaste, SimpleNet
- Pseudo-anomaly generation
- Limited to augmentation strategies

### 4. Vision-Language Models (2023+) ← Our Focus

- Zero-shot detection via text descriptions
- No task-specific training required

# Vision-Language Models - CLIP Foundation



## CLIP (Contrastive Language-Image Pre-training):

### Architecture:

- Image Encoder: Vision Transformer (ViT) or ResNet
- Text Encoder: Transformer for text
- Learned joint embedding space

### Key Capability:

- Aligns images and text in shared semantic space
- Trained on 400M image-text pairs
- Understands relationships: "damaged," "cracked," "normal"

### For Anomaly Detection:

- Compare test image to text descriptions
- "photo of a normal bottle" vs "photo of a damaged bottle"
- Higher similarity score indicates classification

# Methods Analyzed - Overview



## Two Key Approaches:

### 1. WinCLIP (2023)

- Sliding window-based dense feature extraction
- Multi-scale window aggregation ( $2 \times 2$  to  $5 \times 5$ )
- Patch-wise anomaly localization

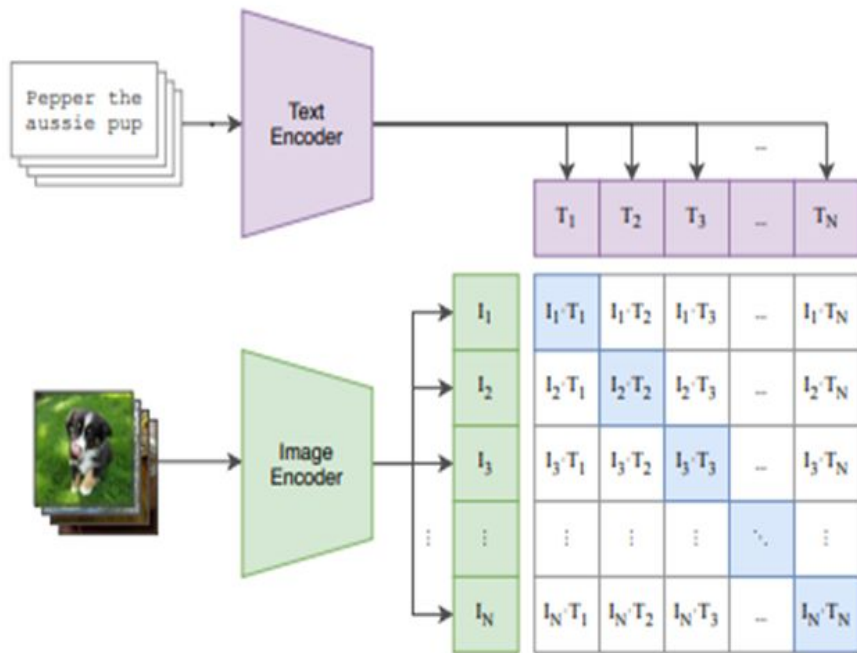
### 2. AnomalyCLIP

- Compositional prompt ensemble strategies
- Enhanced text-visual alignment
- State descriptors: "damaged," "flawed," "broken"

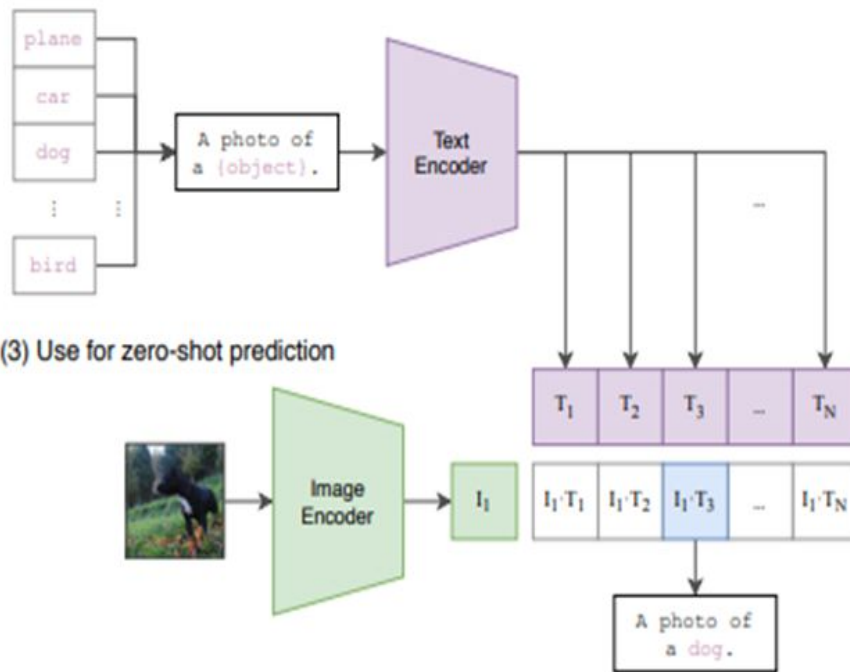


# CLIP Architecture

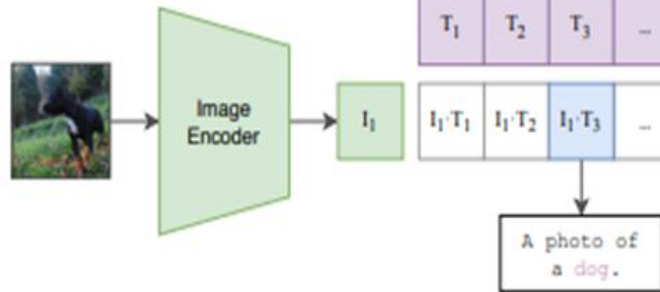
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



# Compositional Prompt Ensemble

## Compositional Prompt Ensemble

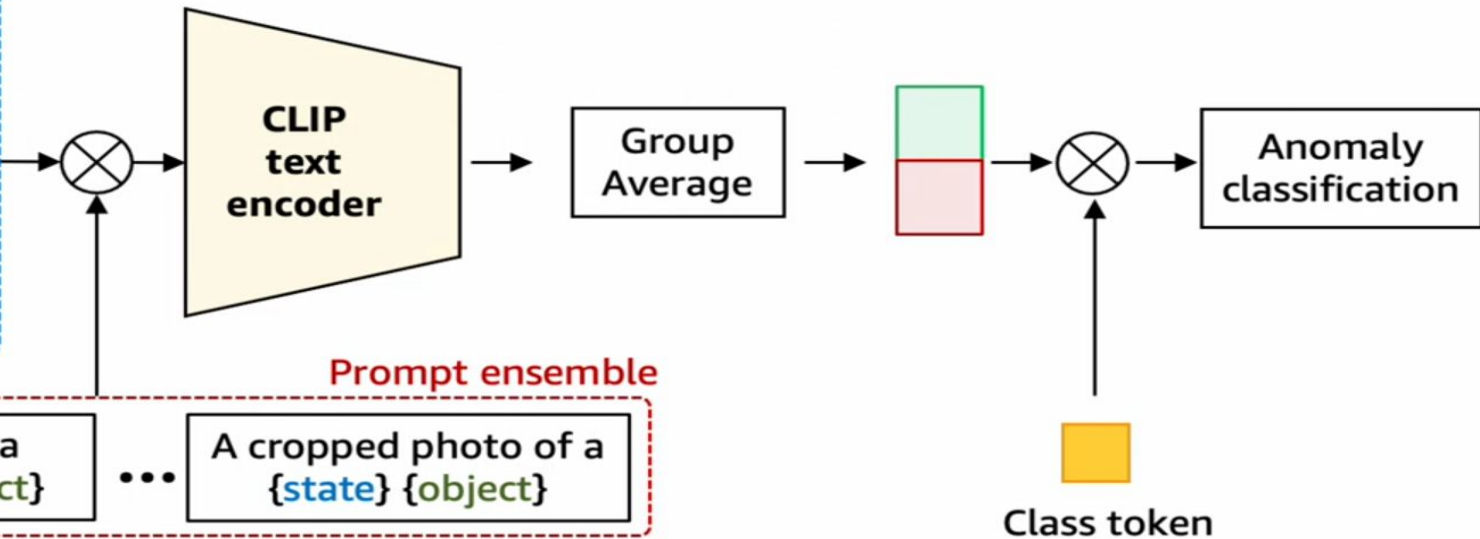
### State ensemble



### Prompt ensemble



ex) object = circuit



# Compositional Prompt Ensemble(AC)



**Two-Class Design:** "normal [object]" vs. "anomalous [object]"

## State-Level Words:

*Normal:* "flawless", "perfect", "unblemished", "[object] without flaw"

*Anomalous:* "damaged", "[object] with defect", "[object] with damage"

## Template-Level Prompts:

"a photo of a [state] for visual inspection"

"a close-up photo of the [state]"

"a cropped photo of a [state]"

**CPE Process:** All combinations → Average embeddings per class → Binary classification

## Why Better Than Naive CLIP?

Naive approach: 74.0% AUROC

**CPE approach: 91.8% AUROC**

# Experimental Setup



## Dataset:

**MVTec AD:** 15 categories (5,354 test images)

- Objects: bottle, cable, capsule, pill, screw, metal\_nut, screw, toothbrush, transistor, zipper
- Textures: carpet, leather, wood, grid, hazelnut, tile, wood

## Evaluation Metrics:

### Classification:

- AUROC (Area Under ROC Curve)
- AP (Average Precision)

### Segmentation:

- Pixel-level AUROC
- AUPRO (Per-Region Overlap)

### Efficiency:

- Inference time per image
- Memory requirements

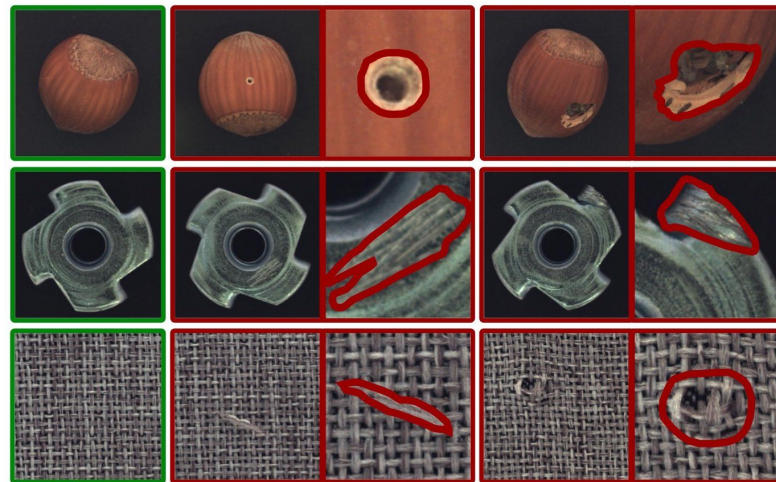
## MVTec AD Dataset Categories

### Defect Types:

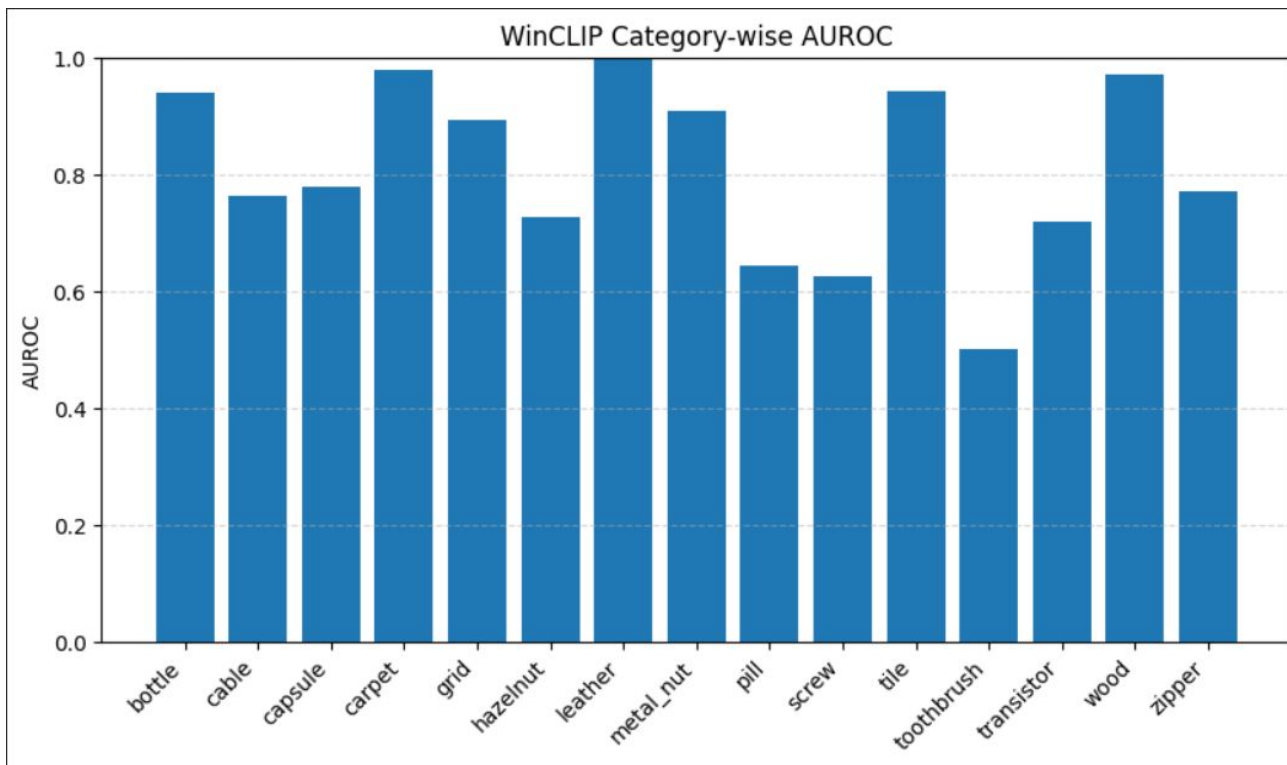
- Surface defects (scratches, cracks)
- Structural anomalies (deformations)
- Logical defects (missing parts, wrong assembly)

### Test Set: 1,725 total test images

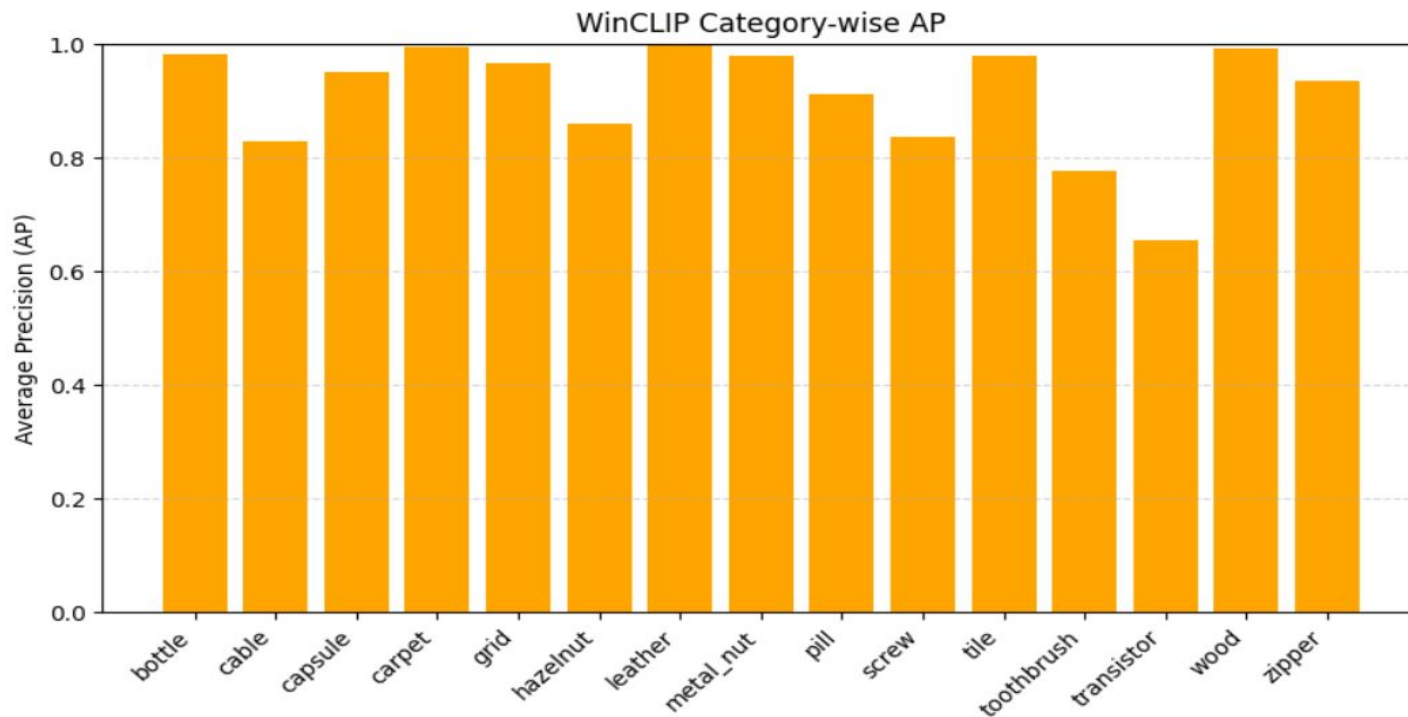
- Mix of normal and anomalous samples
- Ground truth segmentation masks provided

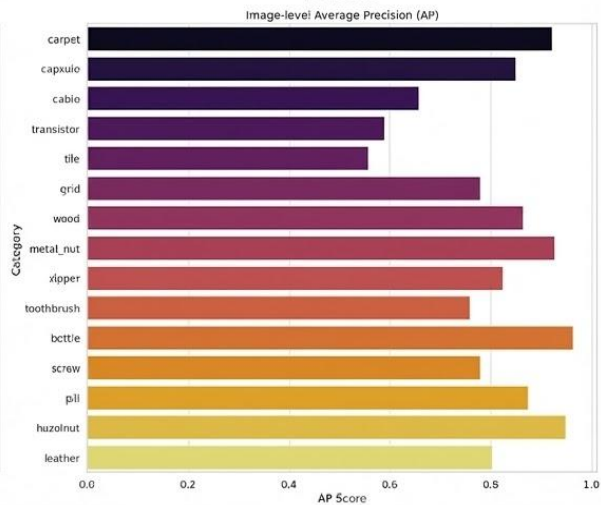
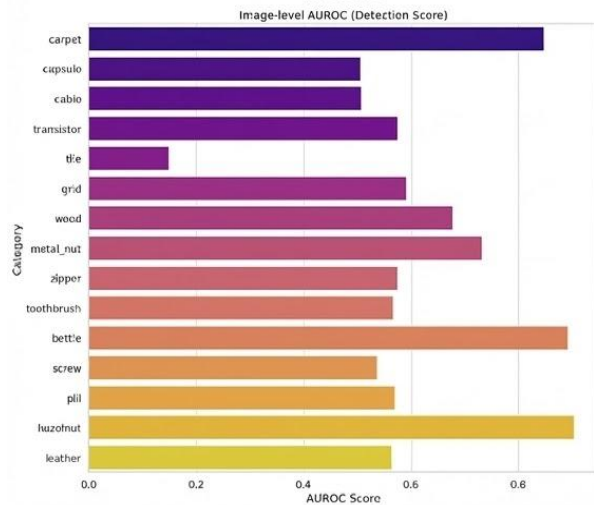
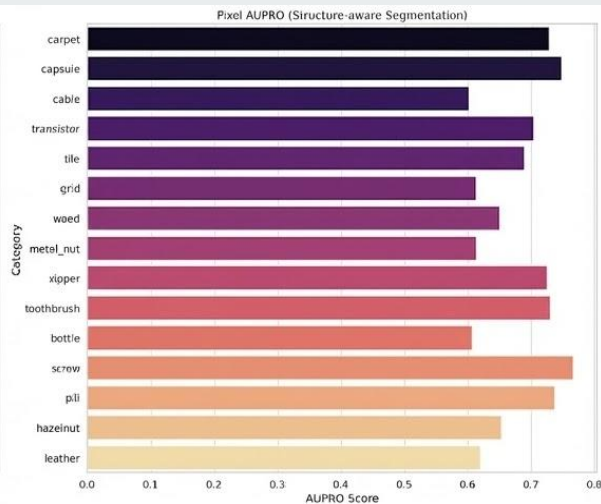
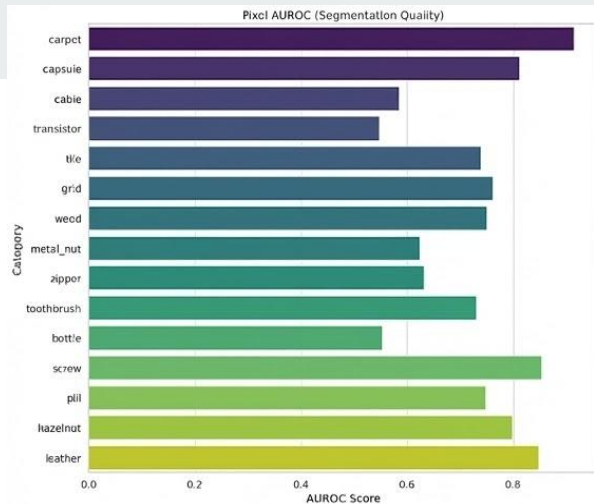


## Anomaly detection (AUROC - MVTec AD)



## Anomaly detection (Average Precision - MVTech AD)





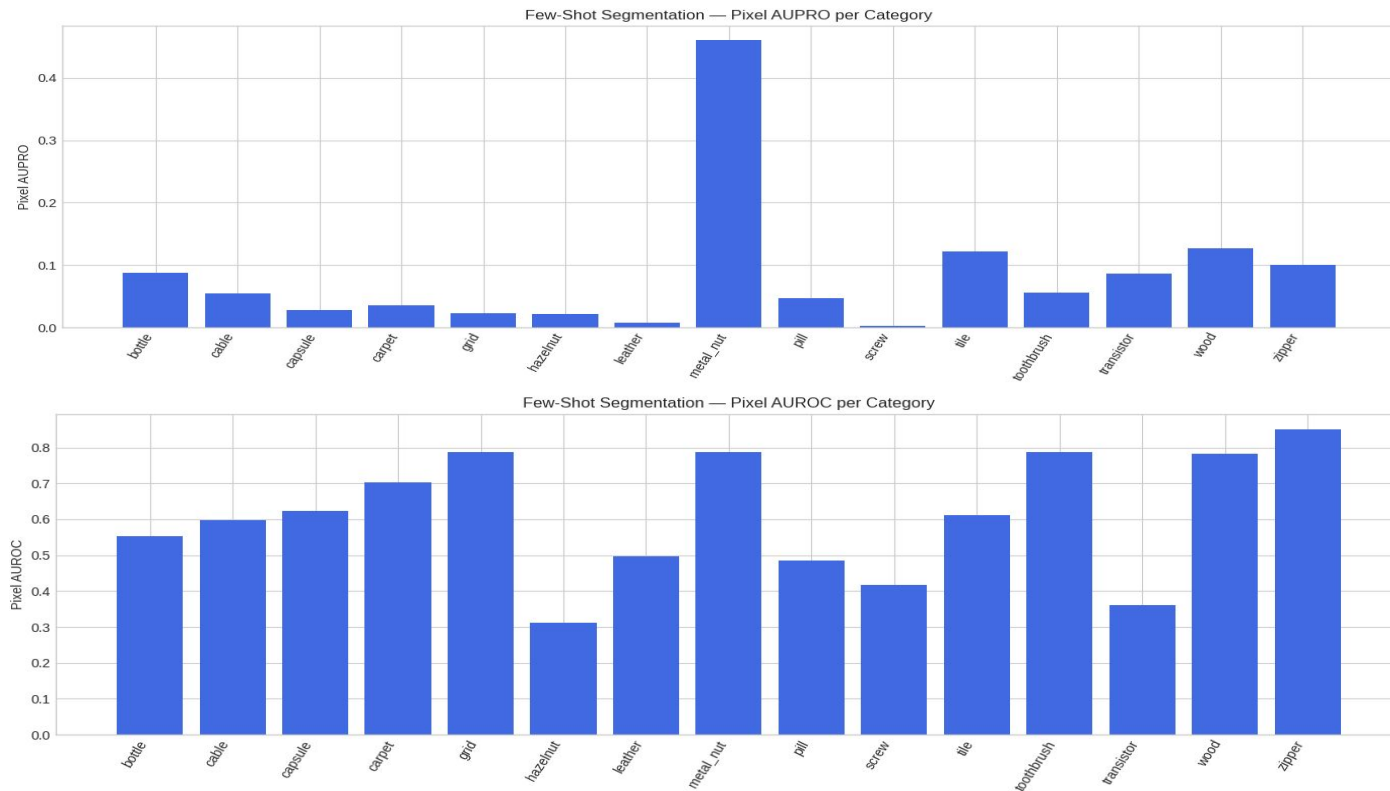
## Anomaly segmentation (Zero-shot)



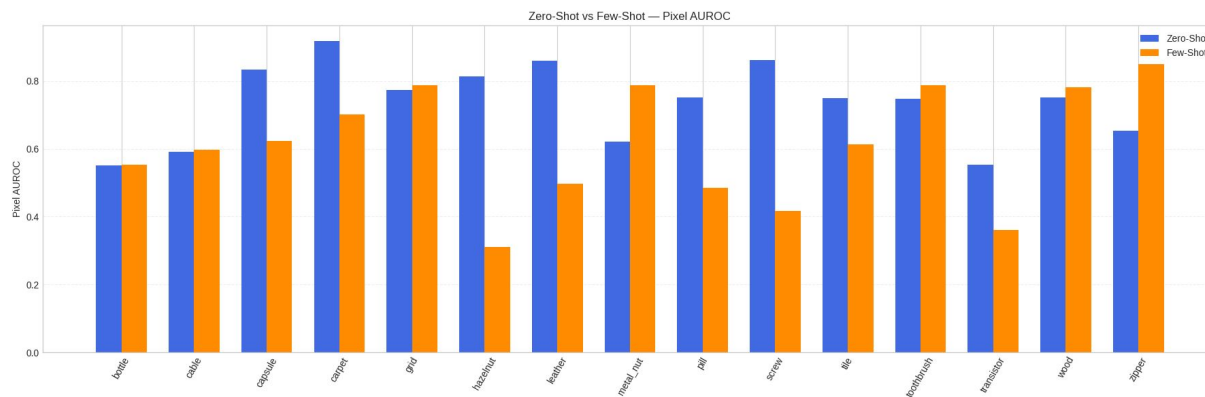
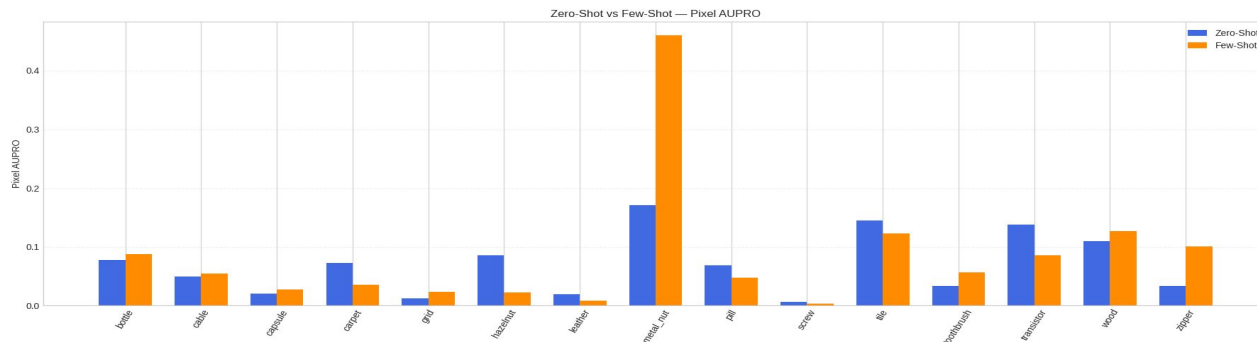
# Anomaly Detection (Few-shot vs Zero-shot) - MVTec AD



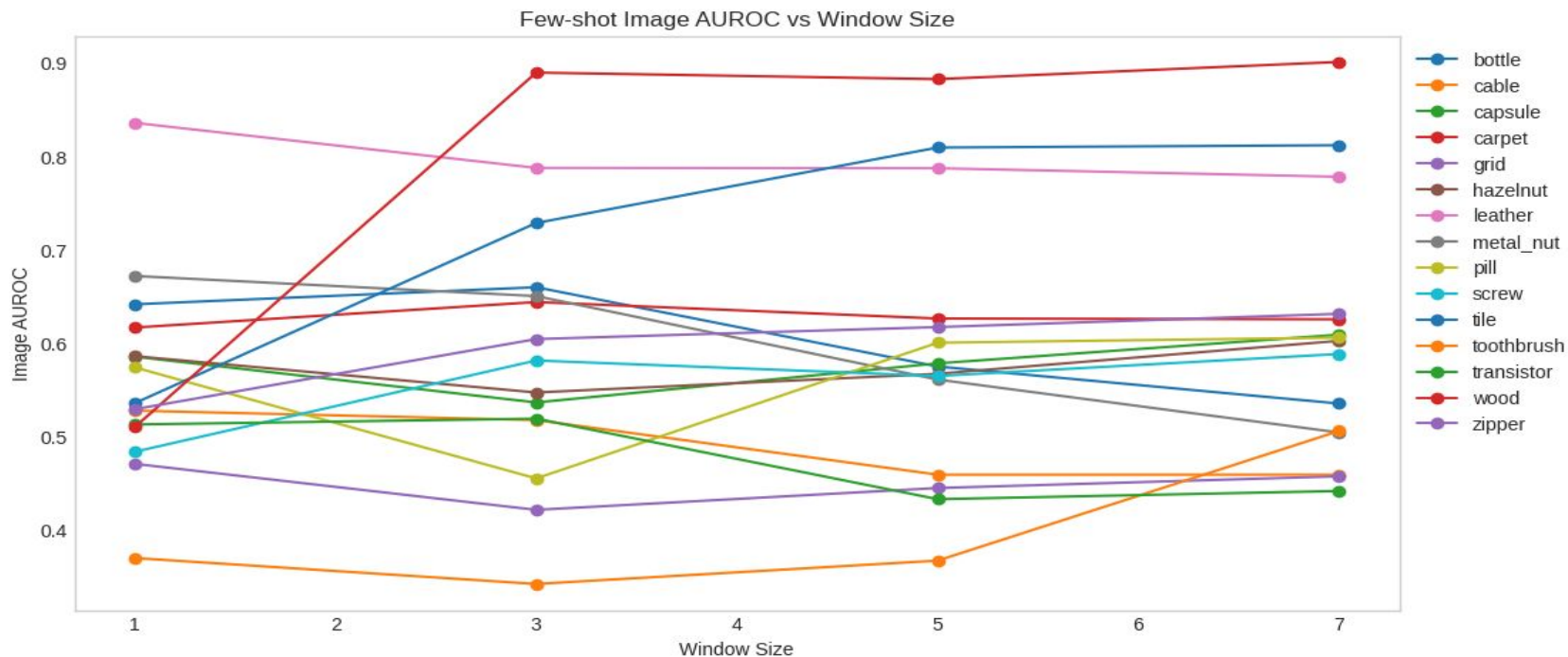
# Anomaly segmentation(Few shot)



# Anomaly segmentation(Zero shot vs Few shot)



# Will changing the window size change the accuracy?



## Results Overview - WinCLIP



- **High zero-shot and few-shot accuracy:** Strong anomaly classification even with minimal supervision.
- **Consistent across categories:** Performs reliably on both textured and structured objects.
- **Segmentation remains challenging:** Pixel-level localization shows limited improvement.
- **Lightweight and scalable:** No retraining required; fast and efficient at inference.
- **Overall:** Excellent for **anomaly classification**, but less effective for **segmentation** tasks.

# AnomalyCLIP - Object Agnostic Prompt Learning



## Building on WinCLIP's approach:

WinCLIP uses sliding windows for local analysis

## AnomalyCLIP takes a different approach:

- Object-agnostic prompts (not object-specific)
- Multi-layer text refinement
- Modified attention mechanism (DPAM)

**Key difference:** Learn *universal* abnormality patterns

(Authors of AnomalyCLIP: Qihang Zhou , Guansong Pang , Yu Tian , Shibo He , Jiming Chen)

# AnomalyCLIP - Object Agnostic Prompt Learning



## Key Insight: Anomalies Share Universal Visual Patterns

*Examples across objects:*

- Scratches: nuts = plates = metal surfaces
- Misplacement: transistors  $\approx$  PCB components
- Tumors: liver  $\approx$  kidney  $\approx$  lung

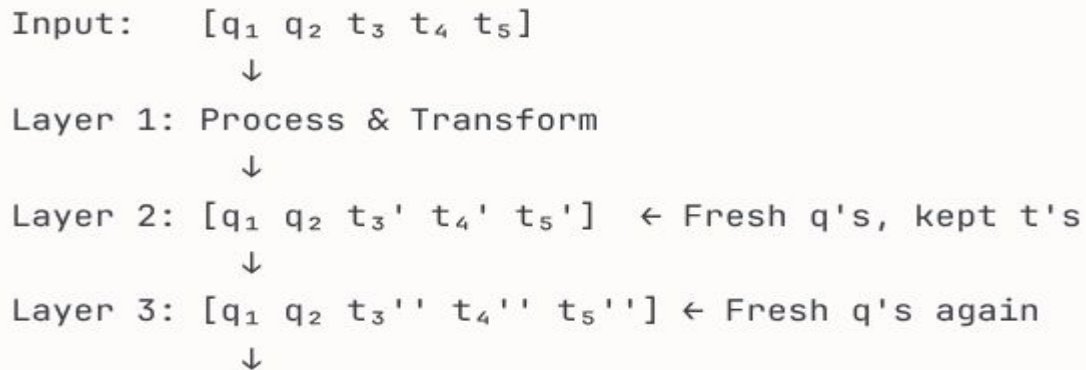
## Strategy:

-  Object-aware: "damaged screw", "defective pill"
-  **Object-agnostic:** "A Photo of  $[V_1 V_2 V_3 \dots]$  Object"

# Progressive Prompt Adaptation Across Layers

**Standard approach:** Learn prompts only at input

**AnomalyCLIP:** Inject learnable tokens at *every layer*



**Why?** Each layer specializes for its abstraction level

- Early layers: Low-level patterns, textures
- Deep layers: High-level semantics, concepts



# GLocal Context Optimization

## Training on Two Levels Simultaneously

$$L_{\text{total}} = L_{\text{global}} + \lambda \sum L_{\text{local}}$$

### Global Loss (Image-level):

- "Is this image normal or abnormal?"
- Cross-entropy on whole-image embeddings

### Local Loss (Pixel-level):

$$L_{\text{local}} = \text{Focal} + \text{Dice}(\text{normal}) + \text{Dice}(\text{abnormal})$$

- Focal Loss: Handles class imbalance (anomalies are rare!)
- Dice Loss: Accurate segmentation boundaries

**Result:** Model learns both *detection* and *localization*

# DPAM - Local Visual Refinement

## Diagonally Prominent Attention Maps

### Problem with CLIP's Vision Encoder:

- Attention mechanism mixes global context
- Local anomaly details → overwhelmed by "object" semantics
- Example: "scratch" signal lost in "screw" context

$$\begin{bmatrix} 0.2 & 0.3 & 0.2 & 0.3 \\ 0.3 & 0.2 & 0.3 & 0.2 \\ 0.2 & 0.3 & 0.2 & 0.3 \\ 0.3 & 0.2 & 0.3 & 0.2 \end{bmatrix}$$

$$\begin{bmatrix} 1.7 & 1 & 1 & 1 \\ 1 & 1.7 & 1 & 1 \\ 1 & 1 & 1.7 & 1 \\ 1 & 1 & 1 & 1.7 \end{bmatrix}$$

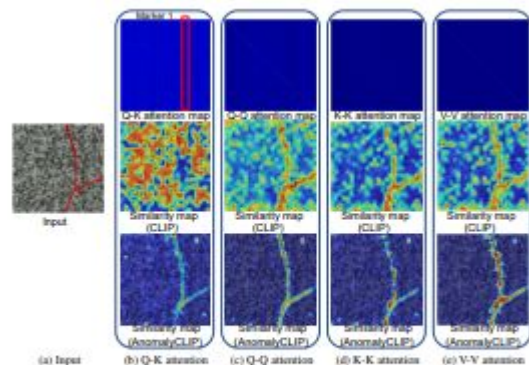
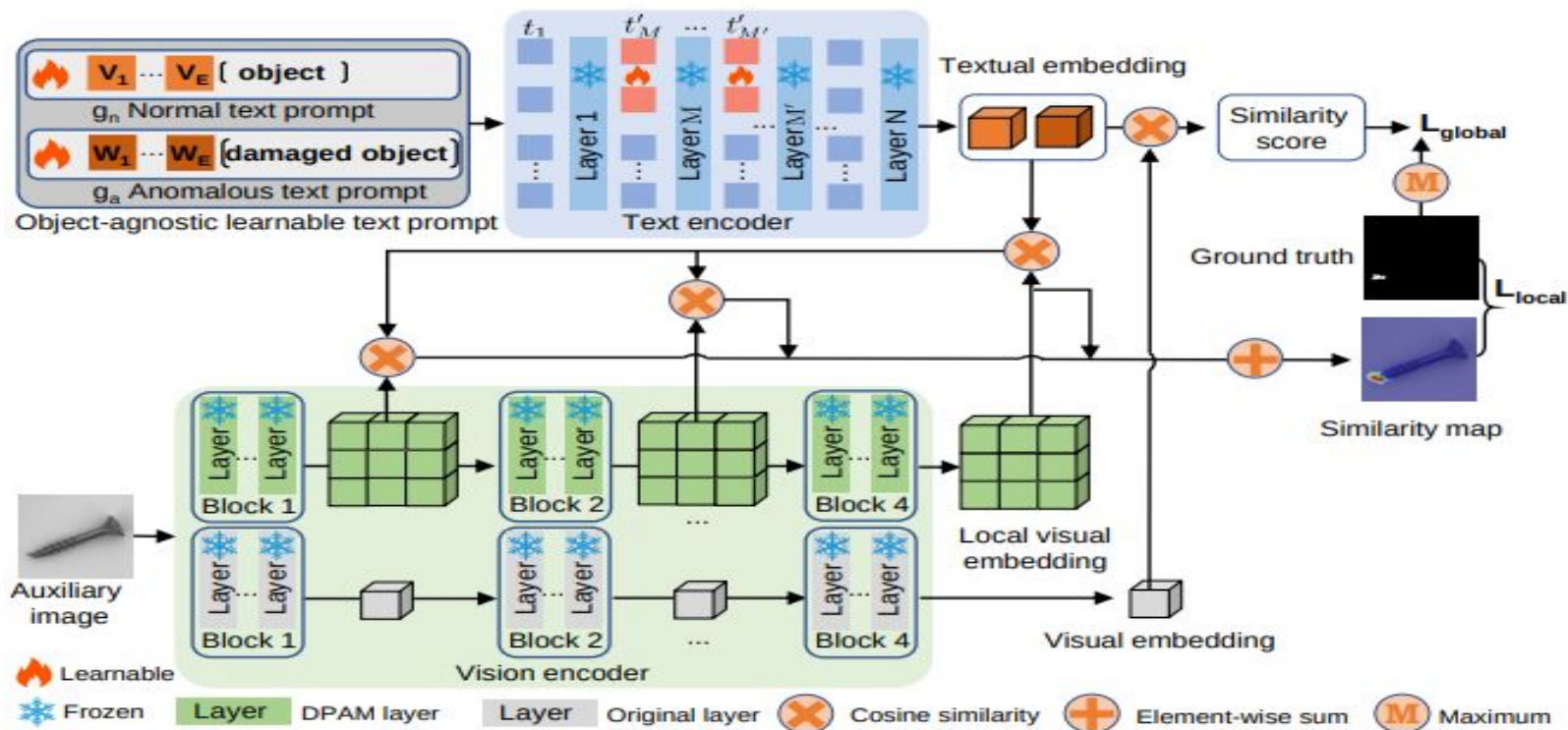


Figure 3: DPAM visualization.

# AnomalyCLIP Architecture



# Putting It All Together

Diverse datasets  
→ Learn universal patterns



Object-agnostic prompts  
+ Multi-layer refinement



Text embeddings:  
Generic normal/abnormal  
concepts  
DPAM visual embeddings:  
Preserved local details



Glocal optimization:  
Image + Pixel level

<- Training

Inference->

New unseen object  
→ Extract features

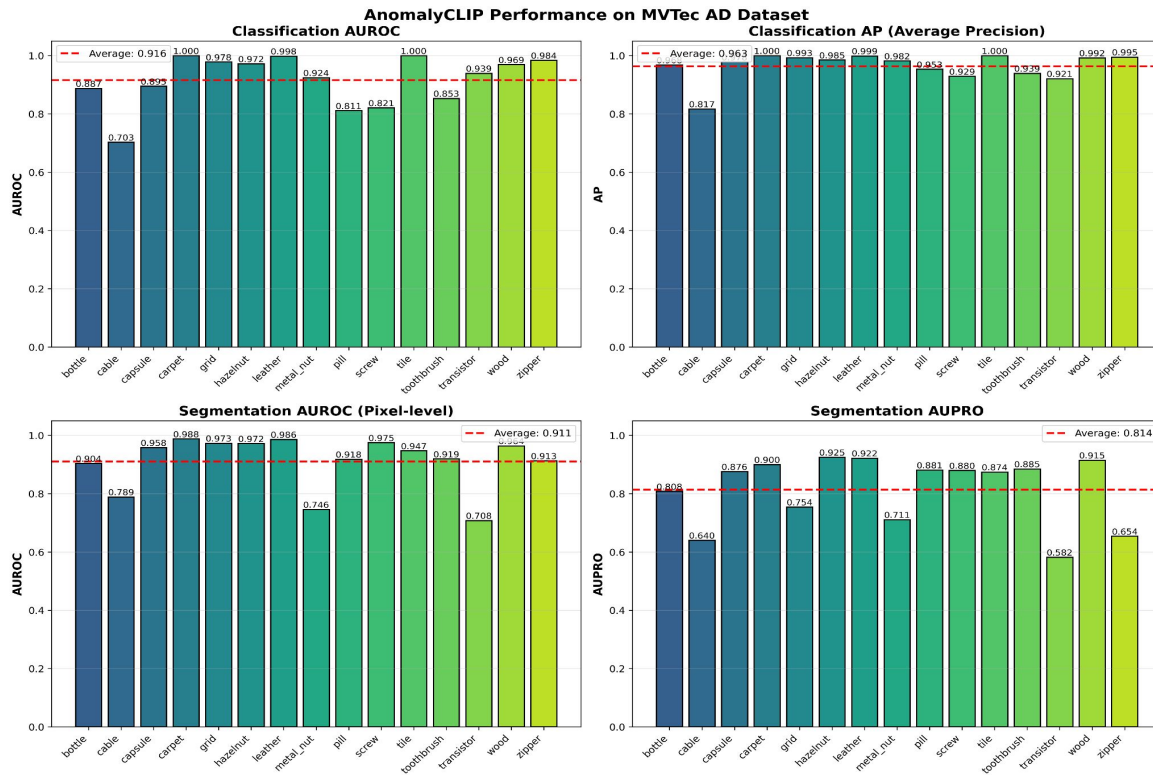


Compare to learned  
normal/abnormal  
embeddings



Output: Anomaly score  
+ Localization map

# AnomalyCLIP Results Overview



# AnomalyCLIP Results Overview(Sub Category)



|          | AUROC | STD   | COUNT |
|----------|-------|-------|-------|
| Contrast |       |       |       |
| Low      | 0.768 | 0.090 | 13    |
| Medium   | 0.900 | 0.098 | 14    |
| High     | 0.939 | 0.101 | 25    |
| Unknown  | 0.973 | 0.050 | 21    |

Performance Defect By Contrast

|         | AUROC | STD   | COUNT |
|---------|-------|-------|-------|
| Size    |       |       |       |
| Low     | 0.861 | 0.126 | 25    |
| Medium  | 0.892 | 0.103 | 14    |
| High    | 0.927 | 0.117 | 13    |
| Unknown | 0.973 | 0.050 | 21    |

Performance Defect By Size

# AnomalyCLIP Results Overview(Sub Category)

| Defect Characteristic | AUROC | Status      |
|-----------------------|-------|-------------|
| HIGH CONTRAST         | 93.9% | ✓ Works     |
| Structural Breaks     | 98.3% | ✓ Works     |
| Large Size            | 89.2% | ✓ Works     |
| LOW CONTRAST          | 76.8% | × Struggles |
| Geometric Deformation | 80.8% | × Struggles |
| Small Size            | 86.1% | ▲ Moderate  |

Discuss only if we have extra time

# Understanding the Results



## Scale for these metrics:

1.0 = Perfect classifier (100% anomalies detected, 0% false alarms)

0.5 = Random guessing (diagonal line)

0.7-0.8 = Acceptable performance

0.8-0.9 = Good performance

>0.9 = Excellent performance

## Area Under the Receiver Operating Characteristic (AUROC)

*How well does the model separate positives from negatives overall?*

- Plots True Positive rate vs False Positive rate, measures global ranking quality



# Understanding the Results



## Area Under Per-Region Overlap curve (AUPRO)

How well an anomaly-segmentation model detects *entire anomalous regions* without over-marking normal areas?

- Measured by area under the PRO curve (Per-Region Overlap vs FPR)

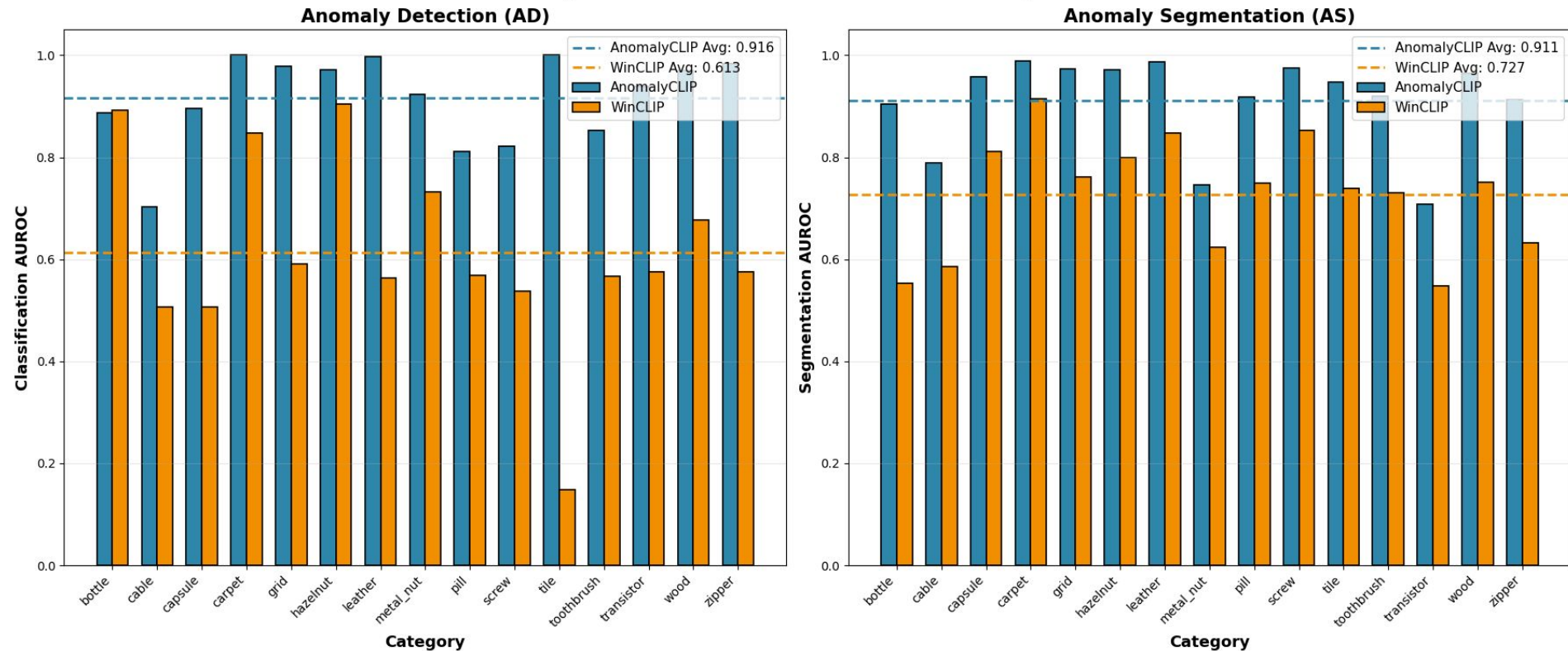
## Average Precision (AP)

How well the model retrieves positive examples when we care about precision and recall?

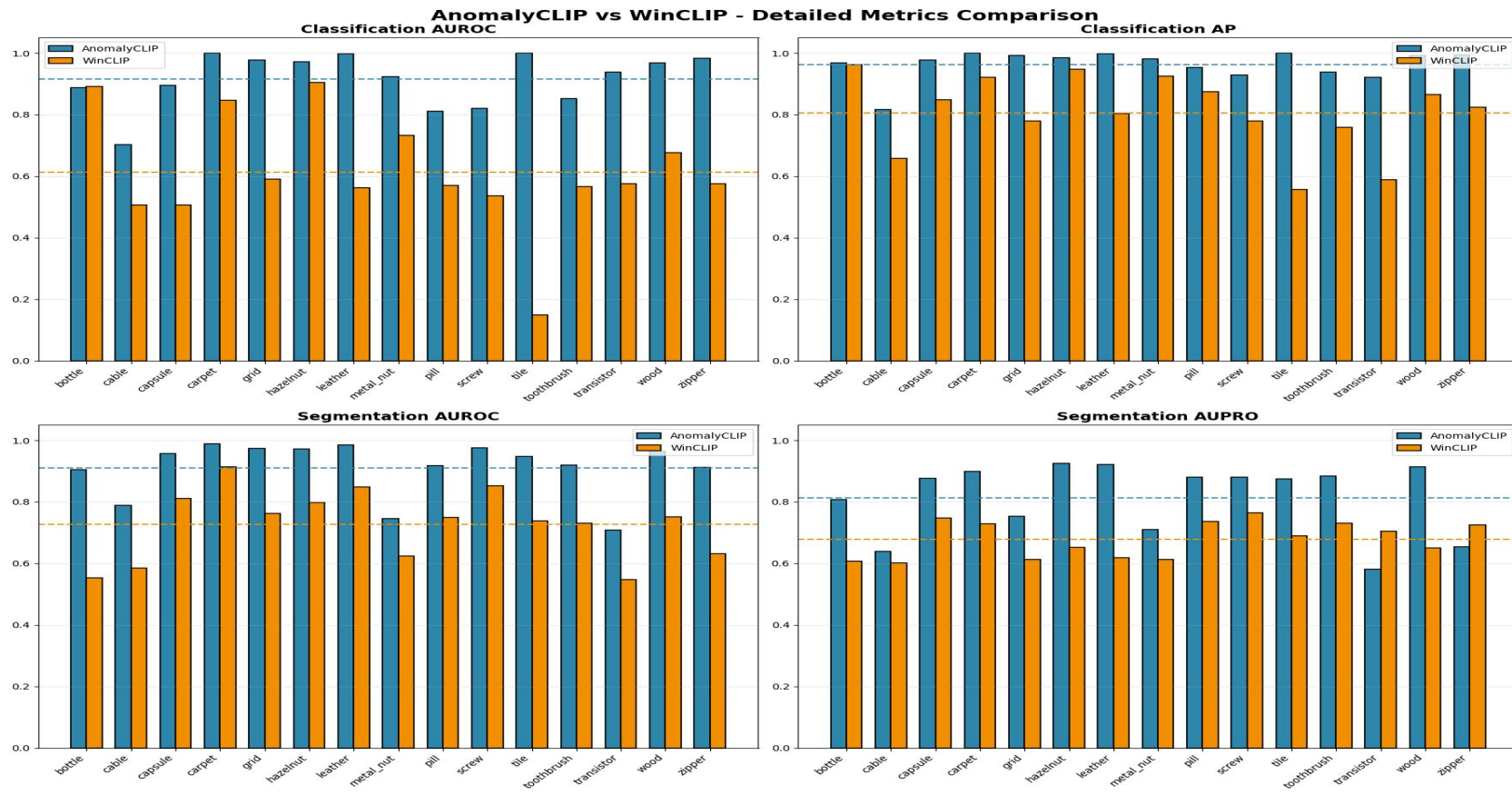
- Area under the precision-recall curve, computed as a weighted mean of precision at different recall levels.

# Detection & Segmentation performance - AnomalyCLIP vs WinCLIP

**AnomalyCLIP vs WinCLIP - Performance Comparison**



# AUROC, AUPRO, AP - AnomalyCLIP vs WinCLIP



# Comparative Analysis - AnomalyCLIP vs WinCLIP



Key Performance Differences:

## **AnomalyCLIP Wins:**

- Better average Classification AUROC: 0.916 vs 0.613
- Higher AP scores across most categories
- Stronger on texture-based defects

## **WinCLIP Advantages:**

- Competitive segmentation performance
- Better on specific object categories
- More fine-grained localization

## **Trade-offs:**

- AnomalyCLIP: Better semantic understanding
- WinCLIP: Better spatial localization
- Choice depends on application requirements

## Qualitative Results



The MVTech AD

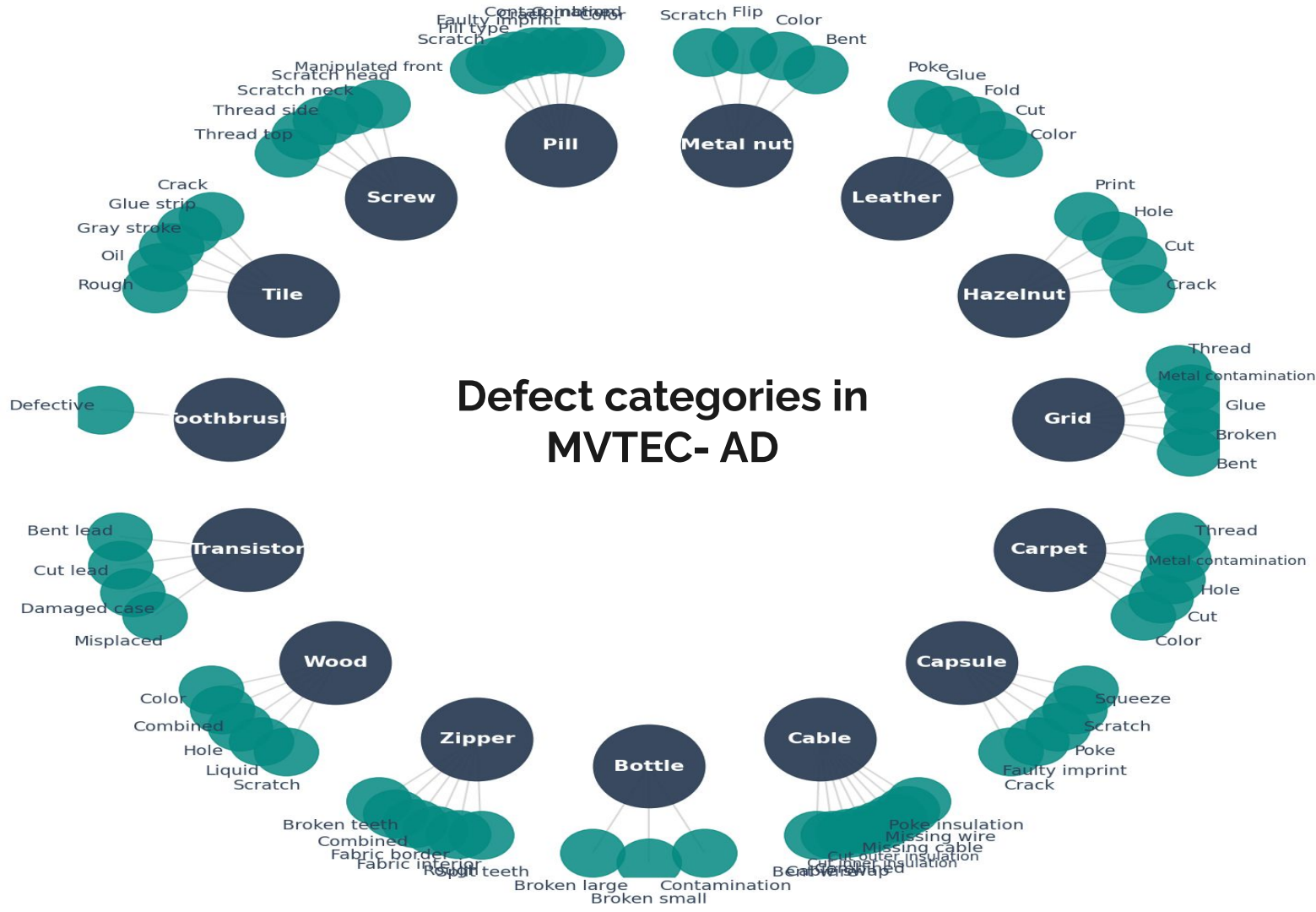
~ 15 different categories

~ 70 sub-categories

For example: A glass bottle has:

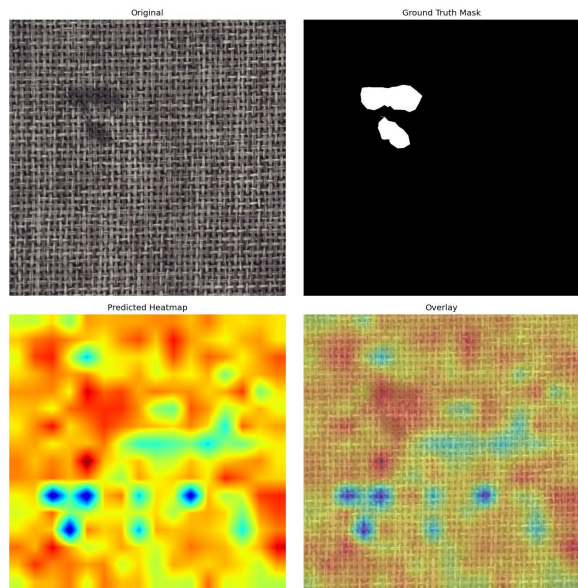
- a. Good condition (no anomaly)
- b. Contamination
- c. Slightly damaged/broken
- d. Heavy damaged/completely broken (maximum anomaly)

## Defect categories in MVTEC-AD

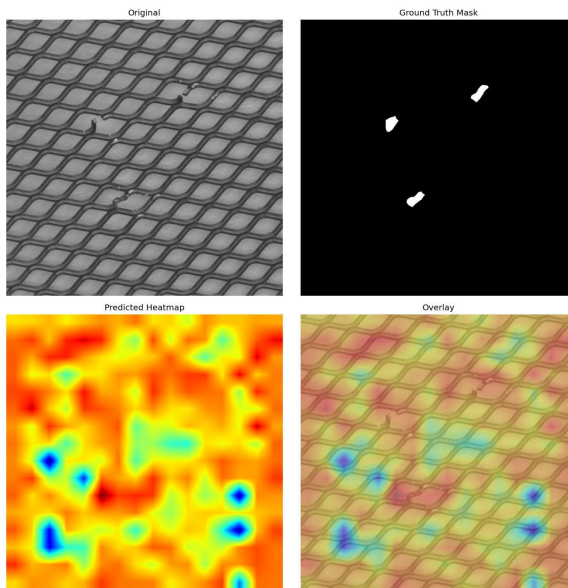


## Qualitative Results

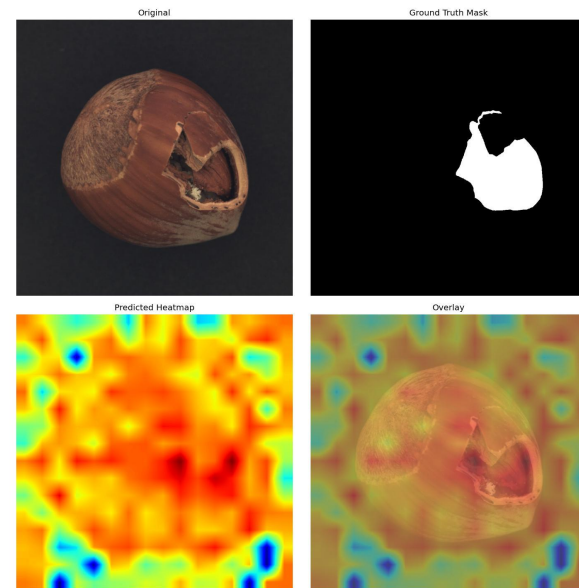
### A carpet with stains (colour anomaly)



### A damaged grid fence



### A cracked hazelnut

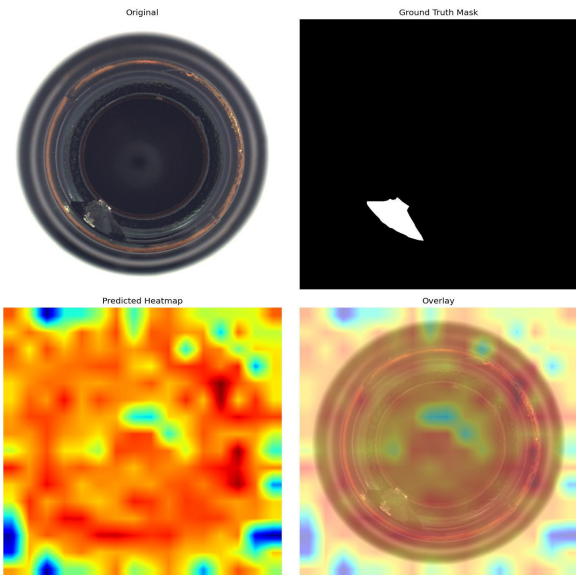




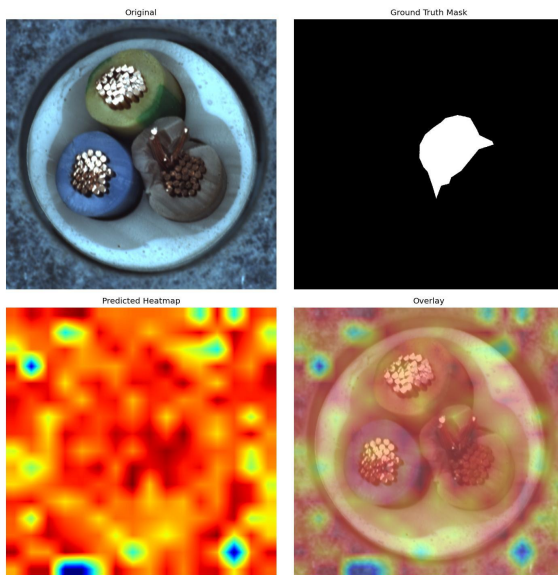
## Qualitative Results



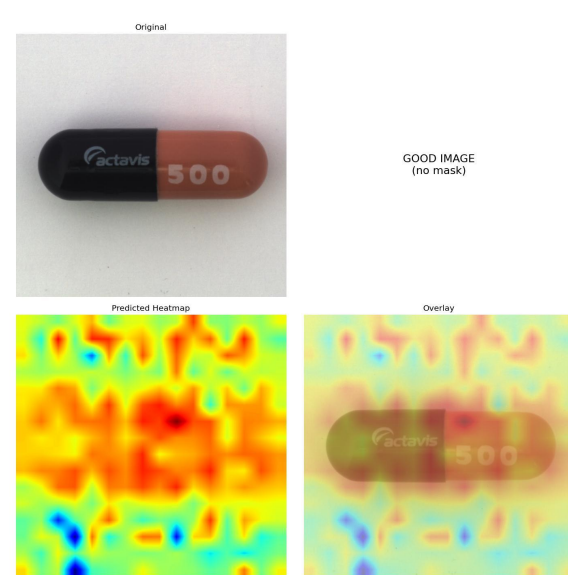
Slightly broken glass  
bottle



A cable with a cut  
inner insulation



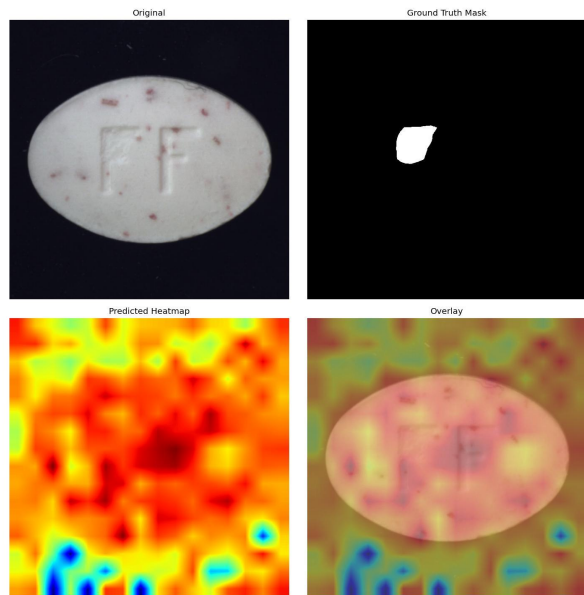
A perfectly fine pill



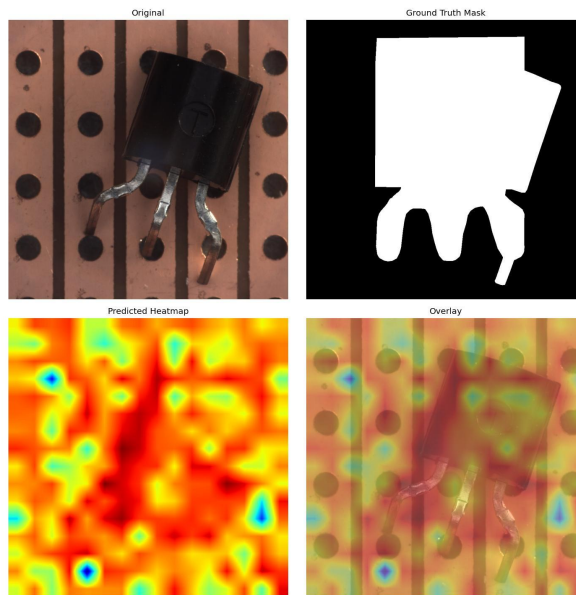


## Qualitative Results

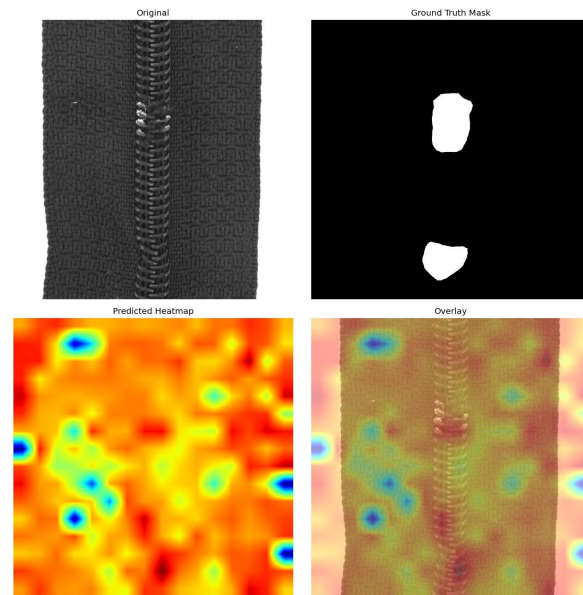
A pill with imperfect imprint



A regular but misplaced transistor on a breadboard



Zipper with 1. Broken teeth and 2. Squeezed teeth



# What Next?



## Our Documentation Focus:

### 1. Comparative Analysis

- Why does AnomalyCLIP outperform WinCLIP?
- Use CLIP-based encoders to analyze differences
- Understand strengths/weaknesses of each approach

### 2. MVTec AD Deep Dive

- Detailed results for all 15 object categories
- Identify performance patterns:
  - Best/worst performing categories
  - Texture vs. structural anomalies
  - Object-agnostic advantages & limitations

**Goal:** Move beyond benchmarks → Understand *why* methods work



# THANK YOU